

Merge Sort

1. Definition

Merge Sort is a **divide-and-conquer sorting algorithm**. It divides the array into smaller parts, sorts them, and then merges the sorted parts.

2. Algorithm

1. Take the elements from the user.
2. Divide the array into two halves.
3. Continue dividing until each part contains one element.
4. Compare elements from the two parts.
5. Merge them in ascending order.
6. Repeat until the complete array is sorted.

Output:

```
Enter number of elements: 5
Enter element 1: 7
Enter element 2: 9
Enter element 3: 5
Enter element 4: 6
Enter element 5: 2
Original array: [77, 9, 5, 6, 2]
Sorted array: [2, 5, 6, 9, 77]
Execution time: 2.7836998924613e-05 seconds

...Program finished with exit code 0
Press ENTER to exit console.
```

Time Complexity

Case	Time Complexity
Best Case	$O(n \log n)$
Average Case	$O(n \log n)$
Worst Case	$O(n \log n)$

6. Space Complexity

$O(n)$

Merge Sort needs extra memory for the temporary arrays used during merging.